of the problem space and managing to make local progresses.

Moreover, it's been around since the 1970s, and that explains why it's one of the favorite heuristics used in optimization; during the last 20-30 years, new optimization techniques have been developed, like *genetic algorithms* (that we'll discuss in the next chapter) and *artificial immune systems* (AIS), which use different biology-inspired approaches to speed-up convergence, but the adoption ratio for simulated annealing remained high, and recently it has also been revived with *quantum annealing*.

Last but not least, you'll improve your ability to plan deliveries over several destination, by learning about the *traveling salesman problem* (aka *traveling salesperson problem*, *TSP* for short) and how to find near-optimal solutions in a reasonable time: continuing our discussion on optimizing logistics for our imaginary e-commerce company, we will move from planning a single trip for each delivery to optimizing the route of a delivery truck across several cities, so that it can be loaded once at the company's warehouse and make several deliveries without going back.

17.1 Simulated Annealing

In chapter 16 we introduced local optimization techniques as a way to improve over mere random sampling algorithms; we discussed gradient descent at length, and alluded to a randomized optimization technique: if you remember our analogy to "marble races", gradient descent would always